CHAPTER 13

Conditional probabilities and binomial distribution

S36 (recalls), S37, S38, S39, S41

Before you start · Conditional probabilities

Without a calculator, in six minutes. Respond briefly. Mobilize trees and events.
choose the targeted review indicated under the questions.

1. Translate " A knowing B ».
2. Write $P(A \cup B)$ where A and B are incompatible.
3. Calculate $P(A \cap B)$ if $P(A) = 0.4$ and $P_A(B) = 0{,}3$.
4. Give the contrary event of "at least 3 successes".
5. What does independence mean? A and B ?
6. Give the expectation of a balanced dice.

Responses

$P_B(A)$; $P(A) + P(B)$; 0,12; not more than 2 hits; $P(A \cap B) = P(A)P(B)$; 3,5.

Targeted review after the diagnostic check

Repeat only the missed questions. For each question: **1.** Name the population after "knowing". **2.** Make a meeting of two disjunct cases. **3.** Multiply the Probability of the branch by conditioning; R46. **4.** List integers below 3 ; R48. **5.** Return to factorization of the probability of intersection. **6.** Summarize the six weighted values questions R are used as support for the teacher, even if their collective ritual is planned later.

Objectives

Build a tree; choose a binomial distribution; calculate a cumulative probability and a threshold; interpret expectation and variance.

Course notes

Reminders: conditioning and independence

If $P(A) > 0$, $P_A(B) = \frac{P(A \cap B)}{P(A)}$. A tree branch carries a conditional probability; probability a path is the product of its branches. For a partition $A_1, \ldots, A_r$, $P(B) = \sum_i P(A_i \cap B)$; if all the $P(A_i) > 0$, it is written $\sum_i P(A_i)P_{A_i}(B)$. Two events are independent when $P(A \cap B) = P(A)P(B)$. For $P(A) > 0$, this is equivalent to $P_A(B) = P(B)$. Not to be confused $P_A(B)$ and $P_B(A)$.

Example. $P(A) = 0{,}4$, $P_A(B) = 0{,}7$, $P_{\overline{A}}(B) = 0{,}2$. Then $P(B) = 0{,}4 \times 0{,}7 + 0{,}6 \times 0{,}2 = 0{,}4$ and $P_B(A) = \frac{0{,}28}{0{,}4} = 0{,}7$.

Bernoulli and binomial

A Bernoulli event has two issues, success of probability p and probability failure $1 - p$. The indicator variable B is 1 if successful and 0 if not: $E(B) = p$, $V(B) = p(1-p)$. A schematic of Bernoulli is a rehearsal of n the same and independent tests. X of success follows binomial distribution $\mathcal{B}(n, p)$.

Law and parameters

For $k \in \{0, \dots, n\}$,

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}.$$

$$E(X) = np, \qquad V(X) = np(1-p), \qquad \sigma(X) = \sqrt{np(1-p)}.$$

Demonstration of the law. Each path comprising k success a probability $p^k(1-p)^{n-k}$ by independence. There is $\binom{n}{k}$ choice of success positions. These paths are incompatible; add up their probabilities.

Cumulative events

$$P(X \leq k) = \sum_{j=0}^{k} P(X = j), \qquad P(X \geq k) = 1 - P(X \leq k-1).$$

$$P(a \leq X \leq b) = P(X \leq b) - P(X \leq a-1).$$

For at least one success, $P(X \geq 1) = 1 - (1-p)^n$. The addition of « at least k » is "strictly less than k ».

Example. For $X \sim \mathcal{B}(10, 0{,}2)$, $E(X) = 2$ and $V(X) = 1{,}6$. $P(X = 2) = 45(0{,}2)^2(0{,}8)^8 \simeq 0{,}3020$ and $P(X \geq 1) = 1 - 0{,}8^{10} \simeq 0{,}8926$. Expectation is neither a certain result nor necessarily the most likely.

Threshold and interval

To look for the smallest n such as $1 - (1-p)^n \geq q$, a loop on n After the logarithm, we can solve $(1-p)^n \leq 1-q$ then round in the right direction. To find the smallest integer k such that $P(X > k) \leq \alpha$, increase k until $P(X \leq k) \geq 1 - \alpha$, then check the previous row. We can also look for an interval $[a, b]$ containing at least one prescribed probability, without assume that it is unique.

Independent tests, beyond two issues

For independent experiments, the probability of a list of results is the product of probabilities. The universe is the Cartesian product of the universes of each experience. By example, for two independent throws of a balanced dice, the universe contains the 36 pairs (i, j), $1 \leq i, j \leq 6$, each probability $\frac{1}{36}$. The probability that the sum is 5 $\frac{4}{36} = \frac{1}{9}$, because couples in favour are $(1,4), (2,3), (3,2), (4,1)$. In a Bernoulli diagram, we code success by 1 and Failed by 0: the universe is $\{0,1\}^n$. Independence concerns the whole succession; a simple equality between two observed probabilities is not sufficient to prove it.

Build the model before calculation

13.1 From the table to the tree

First use: S36.

Example Solved · Change denominator

Of the 1,000 objects, 600 come from workshop A and 400 from B. A produces 12 defective objects and B in product 20. By noting D the defect, $P(A \cap D) = 12/1000$, $P_A(D) = 12/600 = 0{,}02$ and $P_D(A) = 12/32 = 0.375$. The same number 12 occurs, but the reference world changes.

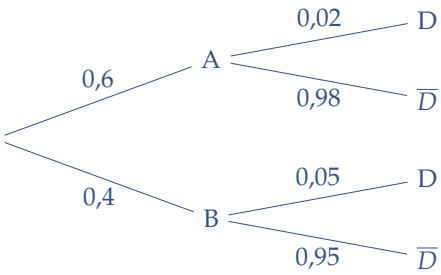


Figure 13.1. Read path A and then D: $0.6 \times 0{,}02 = 0.012$, or 12 items out of 1,000.

A1 · Immediate application Complete the table with two lines A/B and two columns D/no D. Calculate $P(D)$ Are the defect and the workshop independent?

13.2 Find a branch and reverse conditioning

First use: S38.

Ownership · Partition and Total Probability

If A and $\overline{A}$ form a partition, $P(D) = P(A)P_A(D) + P(\overline{A})P_{\overline{A}}(D)$, when the conditional probabilities This result comes from the separate meeting of the two paths leading to D.

Example Solved · Missing Data

We know. $P(A) = 0{,}7$, $P_A(D) = 0{,}1$ and $P(D) = 0{,}13$. Poisons $x = P_{\overline{A}}(D)$. Equation $0{,}13 = 0{,}7 \times 0{,}1 + 0{,}3x$ gives $x = 0{,}2$. The result belongs to $[0;1]$, which is a necessary control.

A2 · Partial guidance With $P(A) = 0{,}4$, $P_A(D) = 0.2$ and $P(D) = 0.14$, find $P_{\overline{A}}(D)$ then $P_D(A)$.

A3 · Refuter "If 90% of patients have a positive test, then 90% of positive people are Build a picture of 1,000 people, with 100 sick, 90 positive among the sick. and 90 positives among non-malads, then calculate the two proportions.

Four characteristics to be checked

First use: S39 and S41.

Definition · Schematic of Bernoulli

Repeat a fixed number n independent tests, with two exits and the same probability of success p. The number X of success follows the law $\mathcal{B}(n;p)$: $P(X=k)=\binom{n}{k}p^k(1-p)^{n-k}$ for $0\leq k\leq n$.

Example Solved · Count Paths

For four probability tests 1 /2, achieving exactly two successes corresponds to six paths, each probability 1 /16. Thus $P(X=2)=6/16$. Independence makes it possible to multiply the probabilities of a path; the binomial coefficient counts the investments of success.

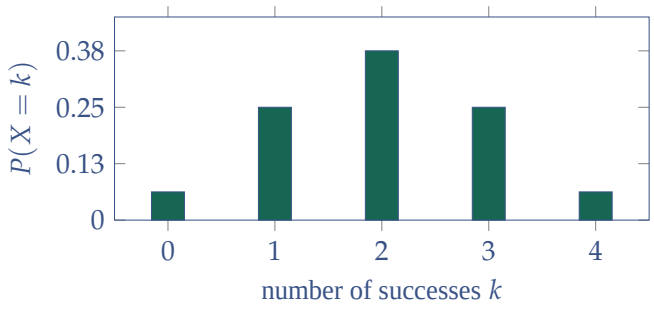


Figure 13.2. Law $\mathcal{B}(4;1/2)$: explain the symmetry and check the sum of the heights.

A4 · Accept or refuse Say in which cases the binomial model is justified: ten independent throws 1 coin; 10 prints without discount in an urn containing five red balls and five blues ; number of launches before the first success. Justify each answer.

13.4 Translate Words into Integer Events

First use: S41.

Example resolved · A loose border of a unit

For X integer, "strictly more than 4" means $X\geq 5$ and its complement is $X\leq 4$. "At least 4" means $X\geq 4$ and its complement is $X\leq 3$. Finally, $P(2\leq X\leq 5)=P(X\leq 5)-P(X\leq 1)$.

A5 · Immediate application For $X\sim\mathcal{B}(8;0{,}3)$, write using cumulative probabilities events: at least 3 hits; strictly less than 3; between 2 and 5 included. Calculate exactly $P(X=0)$.

13.5 Determine a probability rather than a number of tests

First use: S42.

Example resolved · Two tests

For two independent probability trials p, the probability of at least one success is $1-(1-p)^2$. To make it equal to 3/4 is to $(1-p)^2 = 1/4$. Like $1 - p \geq 0$, we keep $1 - p = 1/2$, so p=1/2. The negative sign root would be incompatible with p in [0;1].

A6 · Choice of strategy Three trials are independent, probability of success p. We want the probability of at least two successes of 0 ,9. Express this probability based on p, demonstrate the uniqueness of a solution in $[0;1]$ and frame between two hundredths consecutive.

A first step: In order to be unique, study a function on the $[0;1]$.

Wait a minute at the courthouse. Explain why a probability tree does not involve independence: probabilities on the second level may depend on the first result.

Review postponed · S50 and then S77 Solve a threshold $1 - (1-p)^n \geq c$ by logarithm, then compare exact probabilities and concentration limits.

Research activity

Component control · 40 min.

A component is chosen at random in a production from two workshops. A Event "the component comes from the first workshop", $B = \overline{A}$ and D The event "the component is We're giving. $P(A) = 0{,}6$, $P_A(D) = 0{,}02$ and $P_B(D) = 0{,}05$. The results will be compared Rounded to 10^{-4}.

1. Represent the situation by a weighted tree, then calculate $P(D)$.

2. The chosen component is defective. Calculate the probability that it comes from the first workshop.

3. Twenty components are now selected independently, each following the previous model.
   Note X the number of defective components among the twenty. Justify the law followed by X and specify its parameters.

4. Calculate $P(X = 0)$, $P(X \geq 2)$ and $E(X)$.

5. If the 20 components were taken from the same batch, what hypothesis should the model be?
   Please explain why.

Progressive exercises

E1 · Direct application. Two events are considered A and B such that $P(A) = 0{,}3$, $P_A(B) = 0{,}8$ and $P_{\overline{A}}(B) = 0{,}1$. Calculate $P(B)$then $P_B(A)$.

E2 · Application and justification. Random variable X follows the binomial distribution of parameters 8 and $\frac{1}{4}$. Give the exact values of $P(X = 3)$, $E(X)$ and $V(X)$.

E3 · Method to be organised. Let X a random variable with integer values. k, we're laying $F(k) = P(X \leq k)$. Express $P(X \geq 4)$ and $P(2 \leq X \leq 5)$ with the help of F.

E4 · Choice of strategy. Five cards are drawn successively, without discount, in a set of 52 cards containing four aces. X the number of obtained. The variable X Does she follow a binomial distribution? Justify.

E5 · Synthesis or demonstration. We repeat independently a test whose probability of success is $\frac{1}{2}$. Determine the smallest number of tests for which the probability of obtaining less than 0.9% success.

Work in the manual or annal

Baccalaureate, Métropole, 17 June 2025: Exercise 1, question 5 (p. 3). In class after S41. Read the context of questions 1 to 4; for this work, admit the result $p = 0{,}714$ of question 4. subject is available here: official annal . Use the data and instructions of this yearnal.

Duty at home

DM13 · Reservations and overbooking · 60 min.

A room has eighteen places. The organiser accepts twenty reservations, paid in advance. Each person who has booked presents himself with a probability of 0 ,8 independently of the others. Note X If more than 18 people come, the number of people who come to see each other will increase. surplus persons are refused.

1. Justifying the law of Xthen calculate his expectation and variance.

2. Calculate the probability that the organizer should refuse at least one person. $^{-6}$.

3. Each reservation returns 10 monetary units, including if the person does not come. person refused receives a total payment of 30 monetary units, reimbursement and compensation Understood. $R = \max(X - 18; 0)$ the number of people refused. Express the gain G in function of R, then calculate $E(G)$.

4. Calculate the gain if the organiser accepts only eighteen reservations, with the same rules of Payment. Compare the two strategies from the point of view of average gain.

5. Citing a model hypothesis that can be questionable in practice, then a criterion other than the monetary gain to be taken into account in this decision.

QCM A · conditioning and independence

After S37.

For each question, only one proposal is accurate. consult the correction.

1. $P_A(B)$ means:

A. $P(A \cap B)$ B. $P(B \mid A)$

C. $P(A \mid B)$ D. $P(\overline{B})$

2. If $P(A) = 0.4$ and $P_A(B) = 0.5$, then $P(A \cap B)$ is valid for:

A. 0,9 B. 0,2

C. 0,5 D. 0,1

3. Independent events verify:

A. $P(A \cap B) = 0$ B. $P(A \cap B) = P(A) + P(B)$

C. $P(A \cap B) = P(A)P(B)$ D. $A = B$

4. "Up to 3 hits" means:

A. $X < 3$ B. $X \leq 3$

C. $X \geq 3$ D. $X > 3$

5. In a weighted tree, the probability of a path is:

A. the sum of its branches B. the product of its branches

C. Always $\frac{1}{2}$ D. a difference

QCM B · binomial distribution and threshold

After S41 ; whole counting variables.

For each question, only one proposal is accurate.
consult the correction.

6. $X \sim \mathcal{B}(10; 0{,}3)$ account:

A. the success of ten identical and independent tests B. Ten probabilities

C. Failures only D. a continuous average

7. For $X \sim \mathcal{B}(n; p)$, $E(X)$ is valid for:

A. p^n B. np

C. $n + p$ D. $n(1 - p)$

8. "Strictly more than 4" means:

A. $X \geq 4$ B. $X \leq 4$

C. $X \geq 5$ D. $X > 5$

9. Not all the assumptions in Bernoulli's scheme are satisfied if:

A. there is a fixed number of tests B. Success has the same probability

C. The tests are dependent D. Outcome is success/failure

10. To check a minimum integer threshold in a monotonous problem, we also check:

A. Previous ranking B. A tree colour

C. $P(X = 0)$ only D. the name of X

True or false · expected justification. For each statement, indicate whether it is true or false, and then write evidence or a counter-example.

1. $P_A(B) = P_B(A)$ in any situation.

2. If $X \sim \mathcal{B}(n; p)$, then $P(X \geq k) = 1 - P(X \leq k - 1)$.

3. An observed frequency is exactly equal to the theoretical probability.

Review review

I know how to build a tree, choose or refuse the binomial model and translate an integer threshold. **To be monitored:** Invert conditioning and intersection; forget the shift of an entire bound.

Corrected MCQs

1. **B.** $P_A(B)$ is the probability of B knowing A, for $P(A) > 0$. Intersection and conditioning Inverse are different.
2. **B.** $P(A \cap B) = P(A)P_A(B) = 0{,}4 \times 0{,}5 = 0{,}2$. Successive probabilities are multiplied, they are not multiplied Don't add up.
3. **C.** By definition of independence, $P(A \cap B) = P(A)P(B)$. An empty intersection corresponds to Incompatibility, not independence in general.
4. **B.** "No more than 3" includes 3: $X \leq 3$. "less than 3" would exclude.
5. **B.** The probability of the path is the product of successive conditional probabilities. then disjointed paths to get a meeting.
6. **A.** The number of events is 10, the probability of success 0.3, with independence. counts successes, not probabilities.
7. **B.** A sum of n indicators of success of expectation p has expectation np. The average number failures would be $n(1-p)$.
8. **C.** X is integer: $X > 4$ is equivalent to $X \geq 5$. $X > 5$ wrongly excludes value 5.
9. **C.** Independence is one of Bernoulli's assumptions. His loss no longer allows to automatically apply the model, even if a coincidence of laws may exist in Special cases.
10. **A.** Controlling the candidate rank and the previous one allows to establish the minimality when the probability is monotonous in size. Only one calculation at the candidate rank proves that it is appropriate.

True/false. False: with $P(A) = 1/2$, $P(B) = 1/4$ and $B \subset A$, we have $P_A(B) = 1/2$ and $P_B(A) = 1$. True for X total and k Integer: the two events are complementary. False: a single throw of a coin Balanced has a frequency 0 or 1, not 1/2.

Corrected applications and support

Application 1. Line A: 12 and 588; line B: 20 and 380. $P(D) = 32/1000 = 0{,}032 = 0{,}6 \times 0{,}02 + 0{,}4 \times 0{,}05$. $P_A(D) = 0{,}02 \neq 0.032$: no independence.

Application 2. $0{,}14 = 0{,}4 \times 0{,}2 + 0{,}6x$ gives $x = 0{,}1$. $P_D(A) = 0{,}08/0{,}14 = 4/7$.

Implementation 3. Diseases: 90 positives and 10 negatives; non-malads: 90 positives and 810 negatives. $P_M(+) = 0.9$, but $P_+(M) = 90/180 = 0{,}5$.

Implementation 4. The first one agrees if we have a fixed outcome. Without remission, the probabilities
The number of essays to the first success is not independent.
not fixed: this is not a number of successes in a fixed length scheme.

Application 5. $1 - P(X \leq 2)$; $P(X \leq 2)$; $P(X \leq 5) - P(X \leq 1)$. $P(X = 0) = (7/10)^8 = 0{,}05764801$.

Implementation 6. $q(p) = 3p^2(1-p) + p^3 = 3p^2 - 2p^3$. $q' = 6p(1-p) > 0$ on $]0;1[$, therefore q is strictly increasing on $[0;1]$. Continuity and $q(0) = 0 < 0{,}9 < q(1) = 1$ give existence and uniqueness. $q(0{,}80) = 0{,}896 < 0{,}9 < 0{,}905418 = q(0{,}81)$.

Institutionalization

Keep the table next to the tree; ask at each step "from who?".
integers on an axis before calculating. In parametric problem,
distinguishing the variable X, integer $n = 3$ and parameter p.

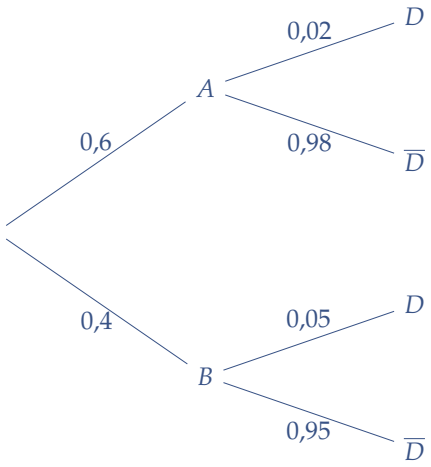
Conduct and correction

Benchmarks for the teacher

Reactivating conditionals in S36 before any Bernoulli schema.
model hypothesis, it does not arise only from repetition. For a draw without discount,
do not automatically invoke a binomial distribution. The formulas of expectation and variance
are used in S39; their evidence in sum of indicators is given in Chapter 14.
requested in December must be able to be obtained by a loop, the logarithm being not yet
taught.

Corrected activity

1. The tree carries the branches A and B, probability 0 ,6 and 0,4. Since A, branches $D, \overline{D}$ carry 0.02, 0.98; since B, they carry 0.05, 0.95.



Total probabilities, $P(D) = P(A)P_A(D) + P(B)P_B(D) = 0{,}012 + 0{,}020 = 0{,}032$.

2. $P(D) > 0$ and $P(A \cap D) = 0{,}6 \times 0{,}02 = 0.012$. Therefore $P_D(A) = \frac{P(A \cap D)}{P(D)} = \frac{0{,}012}{0{,}032} = \frac{3}{8} = 0{,}375$.

3. Each sample is a test with two exits, "failure" or "non-failure", with the the same probability 0.032 of defect. The 20 samples being independent, $X \sim \mathcal{B}(20, 0{,}032)$.

4. $P(X = 0) = 0{,}968^{20} \simeq 0.5218$. For further information,

$$P(X \geq 2) = 1 - P(X = 0) - P(X = 1) = 1 - 0{,}968^{20} - 20(0{,}032)0{,}968^{19} \simeq 0{,}1332.$$

Finally, $E(X) = 20 \times 0{,}032 = 0{,}64$. This expectation does not imply that a sample has a fractional number of defects.

5. A common cause related to the lot could make defects dependent. calculations are based on the independence given in the model; this hypothesis should be reconsidered for another protocol.

E1. Total probabilities, $P(B) = 0{,}3 \times 0{,}8 + 0{,}7 \times 0{,}1 = 0{,}31$. Since $P(B) > 0$, $P_B(A) = \frac{P(A \cap B)}{P(B)} = \frac{0{,}24}{0{,}31} = \frac{24}{31}$.

E2. By the binomial formula, $P(X = 3) = \binom{8}{3}\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^5 = 56\left(\frac{1}{4}\right)^3\left(\frac{3}{4}\right)^5$. Parameters give $E(X) = \frac{8}{4} = 2$ and $V(X) = 8\left(\frac{1}{4}\right)\left(\frac{3}{4}\right) = \frac{3}{2}$.

E3. The supplement to $X \geq 4$ is $X \leq 3$, so $P(X \geq 4) = 1 - F(3)$. For the interval, $X \leq 5$ values $X \leq 1$, therefore $P(2 \leq X \leq 5) = F(5) - F(1)$.

E4. No, in a usual game containing four aces. Without discount, get an ace in the first draw changes the probability of obtaining one in the second: the tests are not independent.
d·as can never be worth 5, while a binomial of parameters 5 and 0 $< p < 1$ gives a Positive probability at this value.

E5. The probability of at least one success is 1 $-(\frac{1}{2})^n$. The condition is equivalent to 2 $^n \geq 100$. Like two. $^6 = 64 < 100 \leq 128 = 2^7$, the first integer is 7. $\frac{127}{128} \simeq 0{,}99219$.

Corrected from DM13

1. X counts the success of twenty independent Bernoulli tests of parameter 0 ,8. Therefore $X \sim \mathcal{B}(20, 0{,}8)$, $E(X) = 20 \times 0{,}8 = 16$ and $V(X) = 20 \times 0{,}8 \times 0{,}2 = 3{,}2$.

2. The room lacks seats if $X > 18$, i.e. $X = 19$ or $X = 20$. These events being incompatible,
$$P(X > 18) = 20(0{,}8)^{19}(0{,}2) + (0{,}8)^{20} \simeq 0{,}069175.$$
The risk is about 6.92 %.

3. Number of refusals R equals 0 for $X \leq 18$, 1 for $X = 19$ and 2 for $X = 20$. By definition of expectation,
$$E(R) = P(X = 19) + 2P(X = 20) \simeq 0{,}0807045.$$
Revenues are 20 $\times 10 = 200$ and payments 30 R, therefore $G = 200 - 30R$. Thus $E(G) = 200 - 30E(R) \simeq 197{,}5789$ monetary units. The 30 already include all payments related to a refusal; do not withdraw yet 10.

4. With 18 reservations, no refusal is possible and the revenue is 18 $\times 10 = 180$, what be the number of present. The expectation of the gain with twenty reservations exceeds this amount d-about 17.58 monetary units.

5. The arrivals of people from the same group can be dependent.
and the consequences of a refusal are not summarized by the expectation of gain.
It is not enough to impose a decision alone.

Corrected work in the manual or an anal

The 100 choices are independent, with the same probability $p = 0.0714$. Therefore $X \sim \mathcal{B}(100; 0{,}0714)$.

$$P(X \leq 7) = \sum_{k=0}^{7} \binom{100}{k} (0{,}0714)^k (0{,}9286)^{100-k} \simeq 0{,}577.$$

Finally, $E(X) = 7{,}14$ and $V(X) = 100 \times 0{,}0714 \times 0{,}9286 = 6{,}630204 \simeq 6{,}63$.